Robot Soccer Self-Localization with Active Inference

Charel van Hoof, Noah van der Meer, Martijn Wisse
Delft University of Technology / VDL Robotsports Team

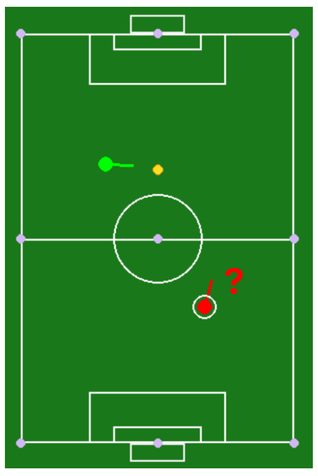
1 RoboCup: the 2050 challenge



2 Self-localization, the idea

Inspired by: Learning grid cells by predictive coding - Prof. Rafal Bogacz

Where am I on the field?



Motion input

Velocity = $\begin{bmatrix} x' \\ y' \\ \theta' \end{bmatrix}_t$ =Odometry Acceleration = $\begin{bmatrix} x'' \\ y'' \\ \theta'' \end{bmatrix}_t$ =IMU

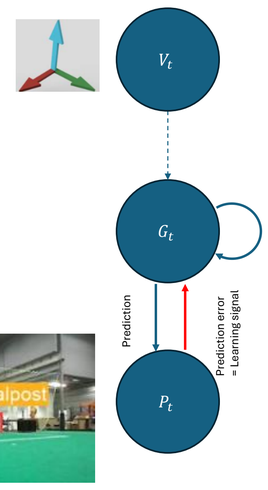
Hidden state - location on the soccer field

$\vec{G}_t = \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{\theta} \end{bmatrix}_t$ In generalized coordinates $\tilde{*} = \begin{bmatrix} * \\ *' \\ *'' \end{bmatrix} = \begin{bmatrix} position \\ speed\ at\ position \\ acceleration\ at\ position \end{bmatrix}$

Observed - landmarks

- predict landmarks
- Observation – prediction = sensory prediction error $\varepsilon_y = \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_\theta \end{bmatrix}$





3 Method

Free-energy minimization by gradient descent

$$\dot{\tilde{\mu}}_g = D\tilde{\mu}_g - D^T\tilde{\Pi}_g\tilde{\varepsilon}_g + \left(\partial_{\tilde{\mu}_g}\tilde{f}(\tilde{\mu}_g)\right)^T\tilde{\Pi}_g\tilde{\varepsilon}_g + \sum_{landmarks}\left(\partial_{\tilde{\mu}_g}\tilde{g}(\tilde{\mu}_g)\right)^T\tilde{\Pi}_p\tilde{\varepsilon}_p$$

Motion prediction errors

Example for x component

- Leveraging generalized coordinates position/speed/acceleration
- $D\tilde{\mu}_{gx} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} \mu_{gx} \\ \mu'_{gx} \\ \mu''_{gx} \end{bmatrix} = \begin{bmatrix} \mu'_{gx} \\ \mu''_{gx} \\ 0 \end{bmatrix}$ *where* $\tilde{\mu}_{gx}$ is the belief of the x in field coordinates (FCS)
- $\tilde{\varepsilon}_{gx} = D\tilde{\mu}_{gx} - \tilde{f}(\tilde{\mu}_g)_x = \begin{bmatrix} \mu'_{gx} \\ \mu''_{gx} \\ 0 \end{bmatrix} - \begin{bmatrix} f_{\mathrm{RCS_2_FCS}}(\mathrm{V}'odometry)_x \\ f_{\mathrm{RCS_2_FCS}}(\mathrm{V}''IMU)_x \\ 0 \end{bmatrix}$
  = motion belief (in FCS) – predicted motion (in FCS)
  - $\tilde{f}(\tilde{\mu}_g) =$ function of motion: estimating the velocity & acceleration, in our case by using the odometry & IMU.
  - Humanoids/Omi wheels can independently move in x, y or θ directions. Hidden state field positions calculations are factorized
- $\tilde{\Pi}_{gx} = \begin{bmatrix} \Pi_{Odo} & 0 & 0 \\ 0 & \Pi_{IMU} & 0 \\ 0 & 0 & 0 \end{bmatrix}$ the precision of the motion prediction errors, in our case the given precision of the Odometry and IMU.
- $\left(\partial_{\tilde{\mu}_g}\tilde{f}(\tilde{\mu}_g)_x\right)^T = 0$ The motion is not dependent of the x position/speed/acceleration on the field location $\tilde{\mu}_{gx}$

Landmark prediction errors

Example for x component

- $\tilde{\varepsilon}_{px} = \tilde{y}_x - \tilde{g}(\tilde{\mu}_g)_x =$ Place cell sensory prediction error = sensory observation landmark – sensory prediction landmark
  - No generalized coordinates, only top row (no sensors for speed/acceleration of landmarks)
  - $g(\mu_g)$ is the translation from robot position (in FCS) to sensory observations (landmarks in RCS)
  - $\tilde{\varepsilon}_{px} = \begin{bmatrix} y_x \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} g_{\mathrm{FCS_2_RCS}}(\mu_g)_x \\ 0 \\ 0 \end{bmatrix}$
    = Observed $_{\text{landmark}}$ (in RCS) – Predicted $_{\text{landmark}}$ (in RCS)
- $\Pi_{px} = \begin{bmatrix} \rho_\ell \cdot \Pi_{camera} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is the precision of the sensory place cell prediction errors
  - Precision of camera (Π_{camera})
  - Option: ρ_ℓ precision distance modifier (closer by observation more precise)
- $\partial_{\tilde{\mu}_g}\tilde{g}(\tilde{\mu}_g)$= Jacobian of $g(\mu_g)$ the matrix of partial derivatives that describes how small changes in the robot state x, y, θ affect the predicted sensory observations of landmarks .

4 Simulator results

The result is not error-free tracking at every instant, but stable recovery: motion terms keep the belief evolving during weak landmark evidence, and later landmark observations correct accumulated deviations.

Table 4: two-hour tracking

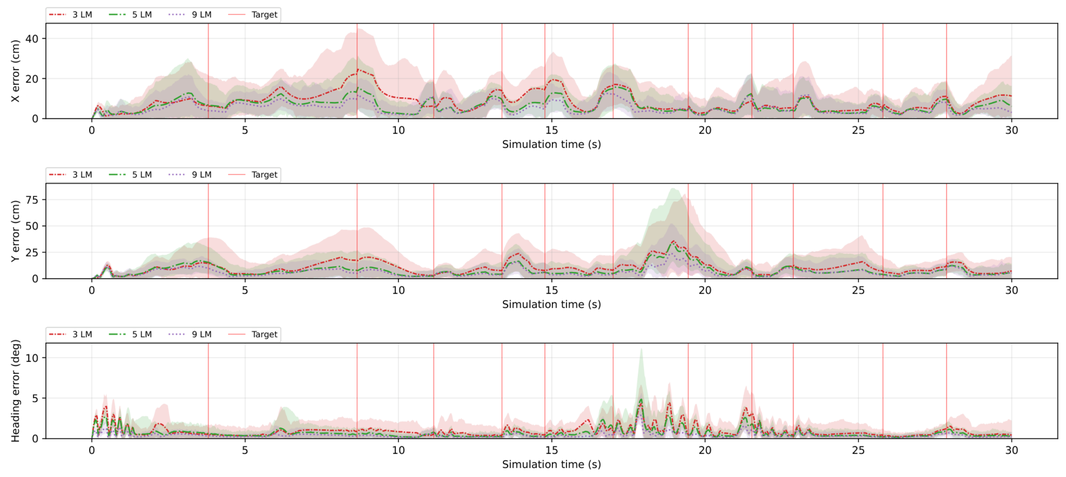
Error threshold	|x|	|y|	|θ|
3 cm / 1°	39.05%	31.37%	94.00%
5 cm / 2°	60.09%	50.21%	98.82%
7.5 cm / 3°	78.71%	68.57%	99.71%
10 cm / 4°	89.69%	81.74%	99.93%
15 cm / 5°	97.67%	94.63%	99.98%
20 cm / 7.5°	99.58%	98.47%	100.00%
25 cm / 10°	99.97%	99.65%	100.00%
30 cm / 15°	100.00%	99.93%	100.00%

- Fast motion: up to 5 m/s and 12 m/s²
- Noise and dropout
- Landmark obstruction (2 s)
- Landmarks ambiguity (e.g. 4 corners)
- 110° field of view

40 Hz, from a near-start belief with gated nearest-neighbor association. Percent of timesteps within each coordinate threshold.

Figure 4: error tracking with 3, 5 or 9 available landmarks

100 matched trials of 30 s per configuration, using unique landmark IDs. Lines: mean error. Shading: 1st–99th percentiles. Red lines: new target.



5 What the results support

Efficient, robust, fast and precise enough for robot soccer in the tested simulations.

Efficient

$$\dot{\tilde{\boldsymbol{\mu}}}_{g_x} = \begin{bmatrix} \mu'_{g_x} \\ \mu''_{g_x} \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ \eta^v_{xy}\Pi_{\mathrm{odo}}\left(\mu'_{g_x} - f(v^{\mathrm{odom}})_x\right) \\ \eta^a_{xy}\Pi_{\mathrm{IMU}}\left(\mu''_{g_x} - f(a^{\mathrm{IMU}})_x\right) \end{bmatrix} + \sum_\ell \begin{bmatrix} \eta^c_{xy}\rho_\ell \boldsymbol{J}^\top_{g_x,\ell}\boldsymbol{\Pi}_{\mathrm{cam}}\left(y_{x,\ell} - g(\mu_g)_{x,\ell}\right) \\ 0 \\ 0 \end{bmatrix}. \quad (12)$$

Approximately 82 µs per step on a modest laptop.

Robust Maintained tracking, as shown in Table 4.

Fast Nearby convergence in tens of steps: means of 23.6 static and 44.2 moving.

Precise Precise enough for RoboCup soccer. Table 4: |x| ≤ 10 cm in 89.69% of steps and |y| ≤ 10 cm in 81.74%.

6 Where it will be used



Test on real robots in match-like conditions

- Is a multi-hypothesis belief needed?
- Use YOLO landmark detection.
- With only one unique landmark (the centre circle), how can we resolve mirror-image pose ambiguity and achieve 100% convergence?
- Benchmark against Kalman-filter localization.